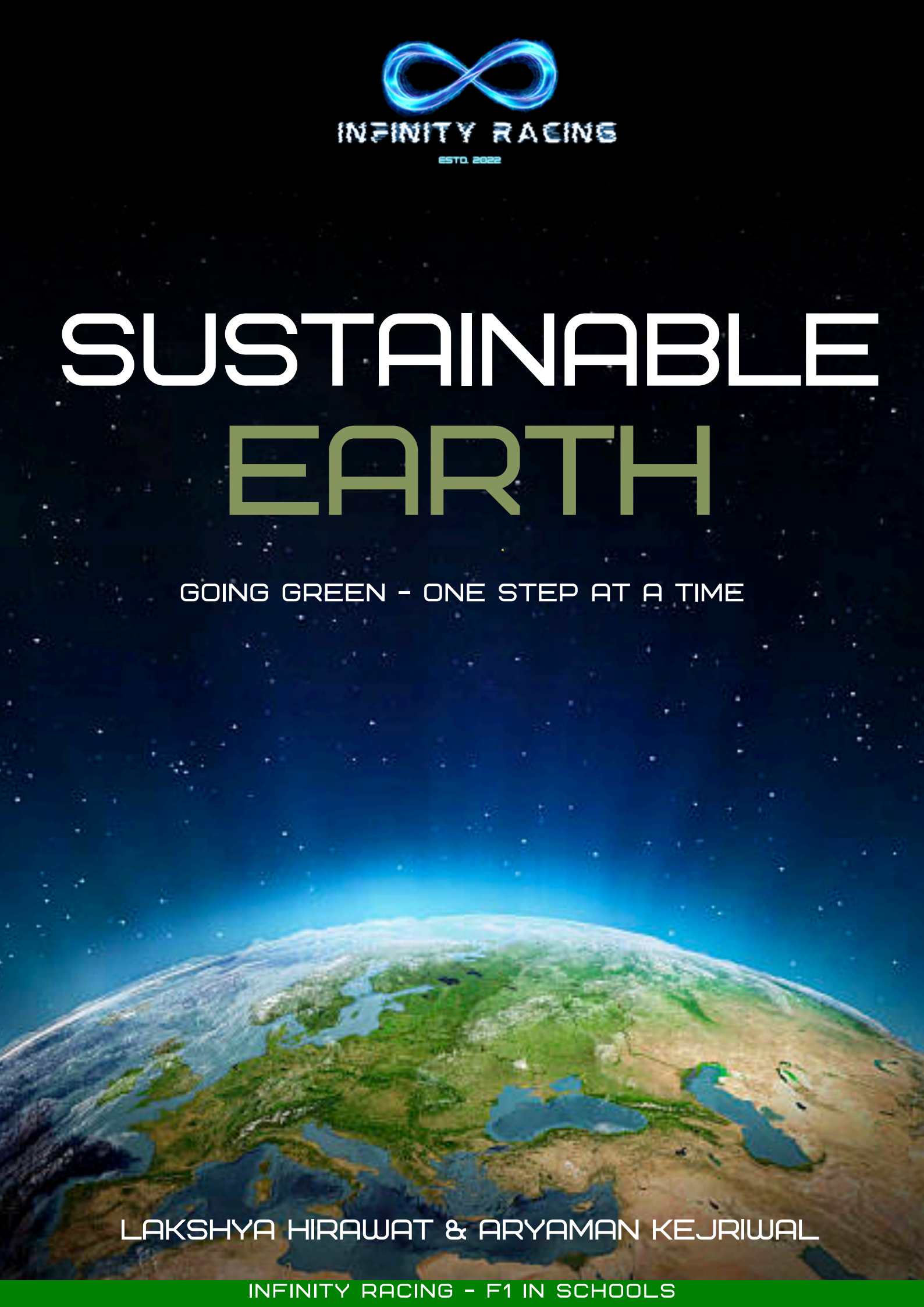
INFINITY RACING

ESTD. 2022

SUSTAINABLE EARTH

GOING GREEN - ONE STEP AT A TIME



LAKSHYA HIRAWAT & ARYAMAN KEJRIWAL

INFINITY RACING - F1 IN SCHOOLS

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TEAM PREFACE

Infinity Racing, a group of six curious teenagers, participating in the F1 In Schools World Finals Competition. We are newcomers in the contest and started our journey on the 24th of March 2022. Each of us reside in Mumbai, India and are from different schools. Infinity Racing will be representing India at the 2023 World Finals and aspire to wave the colours of blue and tangerine on the top of the podium.

Our team consists of young teenagers who are thirsty for knowledge and are extremely interested in STEM (Science, Technology, Engineering Mathematics) as well as Formula 1. Hence, this competition was a complete must-do for us. Our members are:

Lakshya Hirawat (Team Manager, Graphic Designer), Aryaman Kejriwal (Team Manager, Design Engineer), Nathan Doshi (Sponsorship Manager, Manufacturing Engineer), Raghav Sharma (Sponsorship Manager, Graphic Designer), Yuvraj Ruparel (Design Engineer, Manufacturing Engineer) and Rajvir Sehgal (Resource Manager, Graphic Designer).

While being very different and unique from each other, we all share one common goal. Being crowned the 2023 WORLD CHAMPIONS.

MEET OUR TEAM



Lakshya Hirawat
Team Manager
Graphic Designer



Aryaman Kejriwal
Team Manager
Design Engineer



Saniddhya Jain
Team Manager
Design Engineer



Nathan Doshi
Sponsorship Manager
Manufacturing Engineer

WHAT IS F1 IN SCHOOLS?

F1 In Schools is a global STEM competition in which students from countries all over the world compete to ultimately win the World Finals and be crowned the 'World Champions', a goal that every team has set. Each team consists of 3-6 members, each assigned a role which in a way symbolises the working of an actual F1 team. Each team must design a car using the official F1 model block using either CAM or CAD software, showcasing their skill in STEM.

Aside from designing the car the team also has to prepare various documents about the team's journey and design of the car in the form of two portfolios - the Project Management & Enterprise Portfolio as well as the Design & Engineering Portfolio. Every team must also follow a standard set of regulations in order to deem their car legal to race.

There is not only the Technical Regulations but also the Competition Regulations.This competition helps to enhance every member's designing skills as well as their ability to work in an enterprise.

OUR ACHIEVEMENTS

We take immense pride in all of our team's achievements throughout our F1 In Schools endeavour. Infinity Racing has gone through three tough stages of the competition: The India Regional Qualifiers and the India National Finals. While competing against 55 teams at the Regionals which was certainly a challenge, Infinity Racing bagged two awards, The Best Newcomer Award and The Best Project Management & Enterprise Portfolio Award. Along with these prizes, Infinity Racing was crowned the CHAMPIONS of the India Regional Qualifiers.

At the National Finals, Infinity Racing won a total of 3 awards as well as being nominated for a total of 7 awards. We won the People's Choice Award, The Best National Collaboration Award and lastly the Best Scrutineering Award. As we aspired to come in the top 3, the results were finally announced. Infinity Racing stood 3rd (2nd Runner's Up) at the Nationals and had qualified for the World Finals 2023.

We were now representing India on a global stage, at the World Finals being held at Marina Bay, Singapore for 2023,At the World Finals in Singapore last year we managed to grab the Chair of Judges Recognition of Achievement Award which was presented by Oracle Red Bull Racing.

Wish us the best of luck and support us as we set sail for our next season on this journey: the India National Finals, 2024

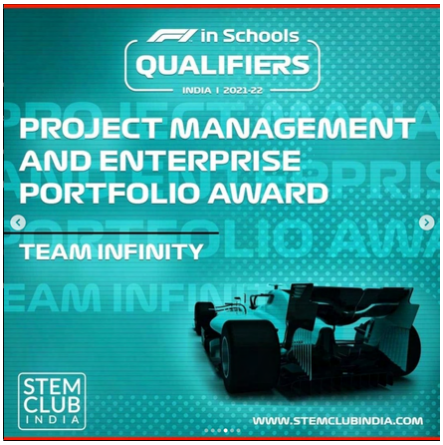
OUR ACHIEVEMENTS

Take a glimpse at some of our prestigious awards through our F1 In Schools endeavour.

India Regional Qualifiers 2022:







National Finals India 2022:











CHAPTER 1 :

A SUSTAINABLE PLANET

Introduction to Sustainability:

Sustainability refers to the idea of maintaining and preserving something of great value. However, the term "great value" does not mean toys, video games or gadgets. Predictions have been made that Planet Earth will end, where will the human race live?

Knowing that, does it not mean that our mother Earth is definitely something of great value and certainly must be preserved? A refined and new definition for sustainability should be the maintaining and preserving of our planet. Sustainability is development that meets the needs of the present without compromising or affecting the ability of future generations and their needs. On our planet we have plenty of resources that help us survive and make life for our generation so much easier such as water and electricity. However, we must understand that these resources are finite and can come to an end. Therefore these resources have to be conserved to ensure there is enough for the coming future generations, without decreasing present quality of life. Hence, much more sustainable choices have to made.

A sustainable society must be socially responsible, focussing on environmental protection and dynamic equilibrium in human and natural eco-systems.



Sustainability generally means avoiding the depletion of natural resources such as fossil fuels and so on, which may harm our future generations in countless ways.

We have mainly thought about our generation, not caring about our future. But now is the time to spread awareness about the various issues that could be caused if we are not careful now.

We must learn how to preserve. All of our resources are not in endless supply and we cannot keep using them like theres no tomorrow. In fact if we do continue this kind of usage, there will be no tomorrow.It is imperative that the younger generations are able to understand the importance of sustainability. Resources are being used left, right and centre. For example, 75% of the electricity generated in India is because of the use of coal, a fossil fuel which is formed over millions of years from decaying remains of animals. It is necessary to understand that if we are so careless now there will be consequences in the not so distant future.

There are multiple ways in which we can preserve these resources.
We can use other means for many of our purposes. Solar energy and Wind energy can be used for multiple purposes. Obviously there will be no end to the sun's rays on Earth and wind will also never stop. It is important that we use these methods at the earliest time possible so that we can help build a sustainable future.

The United Nations also released the Sustainable Development Goals which consist of today's biggest problems we are facing as well as what we must do to preserve our planet for future generations.



Elements of Sustainability:

Sustainability is composed of four extremely important pillars, Environmental Protection, Air, Water and Clean Power.

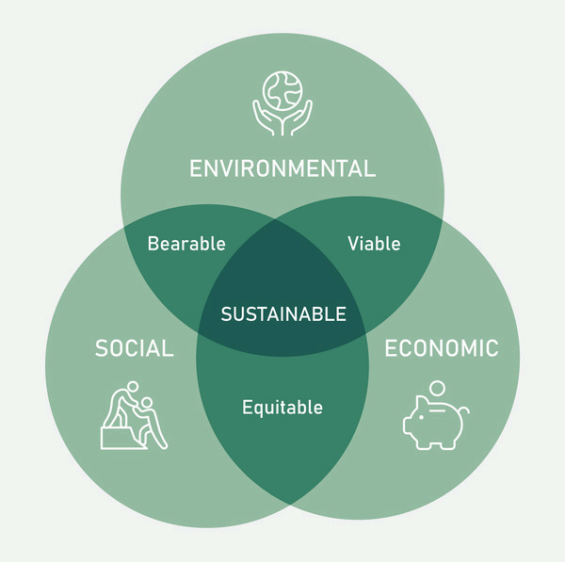
- Environmental Protection: This is concerned with the reduction of Carbon Footprints, water usage, non-decomposable packaging and wasteful processes as part of a supply chain. However, these processes can be cost effective and financially useful as well as important for environmental sustainability.

- Air: This includes regulating air emissions from various sources such as fossil fuels. Furthermore, this concept also authorizes the air quality index. This aids in improving health among people and to reduce hazardous emissions in the air.

- Water: This concept establishes the basic structure for regulating discharges of pollutants into the waters. This also majorly helps in preserving our exquisite marine life and aids in the Sustainable Development Goals (Life below Water).

- Clean Power: This goal aims in using clean power in order to reduce carbon pollution from the power sector. A plan for this was also started called the 'The Clean Power Plan' which will lead to climate and health benefits in our world.

The Venn Diagram shows other elements of Sustainability including environmental, economic and social development. Although, these do not specifically help in sustainability of our planet, these our crucial in our day-today lives.



A more Sustainable Environment:

As said before, sustainability is the action of preserving and maintaining our planet. The Earth includes air, water, soil, and other living and non-living organisms.

To survive on this planet for a longer period of time, we must make our environment and our surrounding much more sustainable! How should we achieve this?

All of us have learnt not to do a numerous amount of things in order to keep our environment sustainable, clean and healthy. However, none of us take these practices into action. This is something we must do at the earliest in order to preserve our environment for the many future generations to come.

A clean and healthy environment is essential for human health and well-being. What we mean by this is, the poor quality of air due to pollution can negatively affect our health. Not only air pollution, but sound and water pollution as well. The things we eat and hear everyday have a direct impact on our health. Hence, to maintain a healthy lifestyle we must also have a healthier and cleaner environment.



Soil and our ground is also an important part of our environment, hence keeping them clean and healthy is also crucial. If we have pollutants in our soil, unhealthy crops will be grown which will ultimately give us food which is not organic. This again has a direct impact on our health and the food we consume everyday.



One of the greatest reasons why we should have a sustainable environment is so we can increase biodiversity and protect our planet's beautiful habitats. Biodiversity is essential for the survival of our ecosystems and animal kingdoms which also includes human beings.



In the 21st Century, mental health is an extremely focused topic and is very crucial for our internal health. Having a healthier, cleaner and sustainable environment also improves mental health. For example, greenery near schools promotes cognitive development in children. Studies have also shown that nature has a link to increased pleasure, well-being and a sense of purpose.



Also another one of the most addressed issue in today's world is climate change and global warming. And the astonishing part is both of these large issues are caused by us, humans. Having a sustainable environment also keeps global warming and climate change minimum. The environmental impact of climate change and global warming cannot be overstated and therefore actions for a healthy and sustainable future have to be done as soon as possible.



A more sustainable environment is one of the most important solutions for many of today's largest problems such as climate change, global warming, human health and medicine. Maintaining and preserving our environment is good for us at the present moment as well as for coming future generations.

To achieve a more sustainable environment, even the smallest tasks like closing the tap and turning off the lights can help make our environment more healthy and clean. Although at our time since these addressed issues are so high worldwide companies as well as environmentalists are running huge campaigns and protests in order to help make our planet a cleaner place to live as well as spread awareness of the greatest problems mankind is facing.

The below pictures show famous campaigns/protests taken place for sustainability and famous Swedish Environmentalist Ms. Greta Thunberg.







A Sustainable Future:



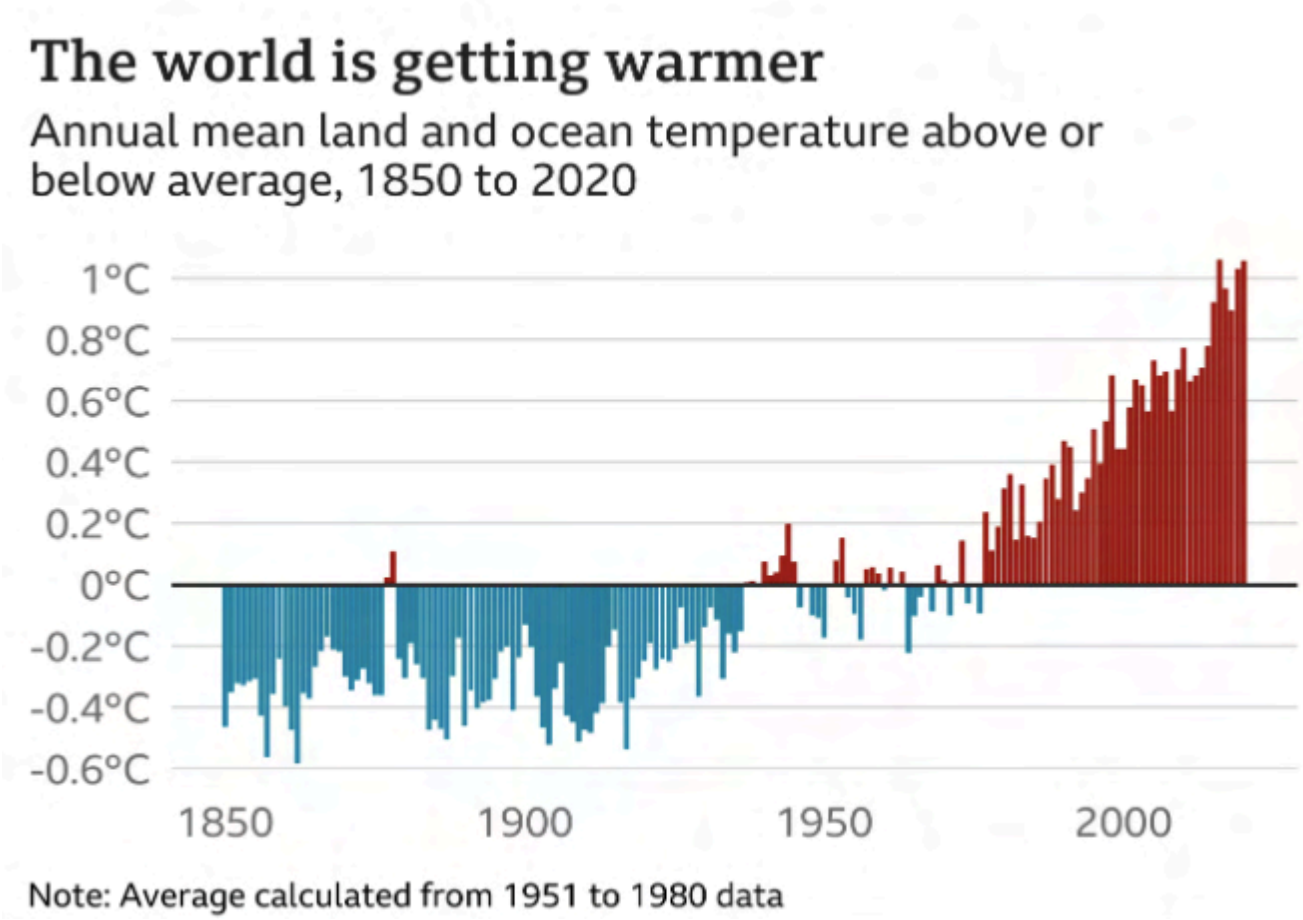
Everyone imagines what the future will be like. Fantasizing about what our world would look like later. But what if I told you that while building a world where cars can fly and human like AI is great, we must also think of a sustainable future. A sustainable future is where we continue to develop our world without compromising the compromising the ability of future generations to meet their own needs.We must take steps that will help to reduce our dependance on non-renewable resources.

But what exactly does a sustainable future look like ?



First off, everyone will have access to clean water and sanitary conditions. Currently, a fifth of the world's population resides in places with a water shortage. There are 785 million people that lack access to adequate water for drinking. However, steps are being taken, and advancement is being made. Significant innovations are being developed, such as the LifeStraw. By removing bacteria, parasites, and other contaminants, this straw may turn murky water into clear, healthy drinking water for about $20. This has the potential to be the answer to the world's water shortage. It has been expected that practically everyone on the earth would have access to safe, non-contaminated water services in the future, between 50 to 100 years from now. Since water is a non-renewable resource such an invention could help out future generations massively as there wouldn't be the need for them to only drink readily purified water.

We've been discussing climate change for roughly a decade or two. As the years passed, it simply continued to grow larger and larger. This occurs as a result of the gradually deteriorating climate. The average temperature of the planet is rising every year, and in 2021 it increased by a surreal 1.28 degrees Celsius. Given where the world is going, more people will become aware of the effects of climate change. There will be a significant increase in the number of people attempting to aid in the prevention of catastrophic conditions emerging. This will put pressure on numerous governments and international organisations to act quickly to combat climate change. This could potentially lead to reduction in the deterioration of the Earth's conditions It will benefit future generations because Earth's conditions are unlikely to get worse.



Important Measures:

We know what you are thinking! We hear so much about people raising awareness and what is happening to our environment, but what are the measures we have to take to preserve our environment? Let's start off with the most basic but most effective way we can heal it, THE 3 Rs.
Reducing, Reusing and Recycling will aid in conserving the tons of harmful daily waste we humans throw away.

- Reduce: Waste reduction can result in improvement in the health sector, due to less toxins being released in the air and water bodies which can transfer diseases from person to person. Useless waste such as lithium batteries which are thrown out from households takes thousands of years to decompose which can harm wildlife as well as the atmosphere.

- Reuse: Reusing waste for different purposes can be beneficial because it lessens the demand for fresh natural resources like wood, oil, textiles, and other materials, reuse offers a good, environmentally friendly alternative to traditional waste management techniques. For example, reusing materials or giving friends or a charity your old clothes.

- Recycle: Recycling is probably the most famous out of the 3 Rs. When you step outdoors there maybe a chance that you have witnessed a recycling bin, where waste products which have the ability to be crafted in to something else useful are placed. Recycling significantly reduces pollution as waste get repurposed for reuse, and can be made into several amazing products!



Alright, since we have revised the positive impacts of the 3 Rs lets talk about another extremely important method we should imply to preserve the environment, education. Yes, it may not seem like much but educating an individual or those who are underprivileged to afford an education, about what mankind does to nature everyday can shock them into changing their lifestyle. A few alarming factors about the environment that made us realise the true damage we have done to the environment are:

- Climate change could be irreversible by 2030.
- More than 1 million species face extinction.
- 78% of marine mammals are at risk of choking on plastic.

Startling, isn't it? Well it is the reality that we humans might regret in later stages of life.
Sharing facts like the ones above can result in more humans taking small but impactful steps which can influence others to do so as well. This is the power of education!

Another strategy of conserving and sustaining our surroundings is purchasing environment friendly products:
If you're in the market for a new vehicle, search for a model that is fuel-efficient. You'll save hundreds on gas and lessen your carbon footprint over time. To identify the most energy-efficient appliances when buying a new refrigerator, washer, or dryer, seek for the Energy Star label. If your household installs a new energy consuming device, consider switching to a clean energy source such as solar or wind to power it.

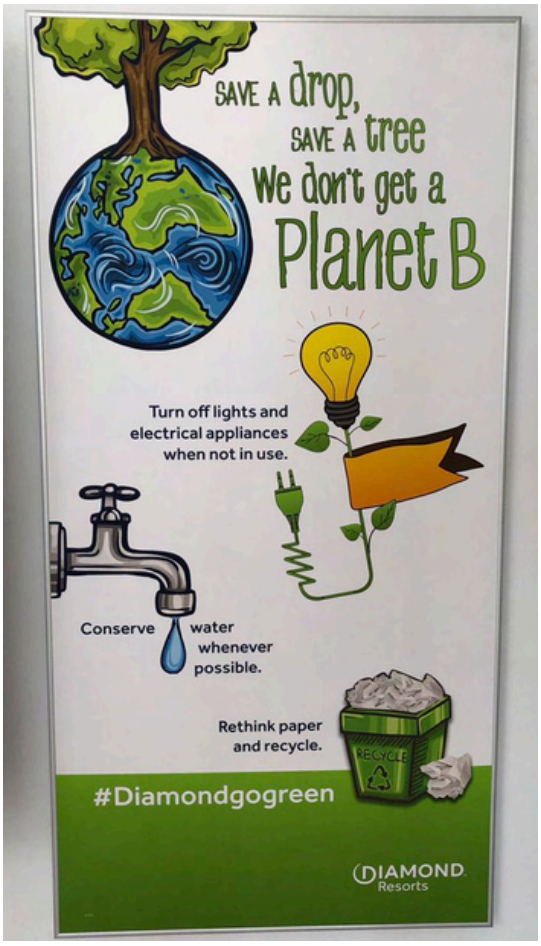


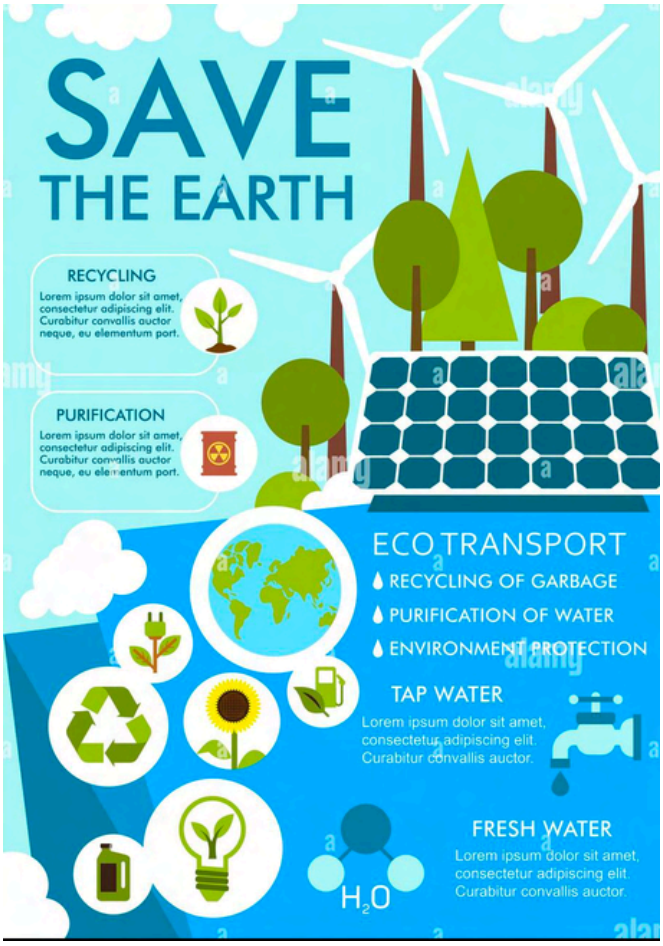


Contribute to the Planet:

Now that you know the important measures that are taken in order to keep our planet sustainable, find out how you can contribute to the planet! Below are some of the simplest ways you can make your mark in sustainability for a greener, healthier and cleaner future.

If you are a young student studying in your school, SPREAD AWARENESS! Spreading awareness on situations like climate change and global warming in your school can help in numerous ways possible. Letting people know about the challenges are planet is facing today is crucial so the word can spread. Spreading Awareness can be done in the form of posters, flyers, webinars and many more ways. Use your creative imagination and make artistic poster and flyers. Make sure to include some of the important measures we have mentioned in the previous chapter. Take a look at some beautiful posters made below. (Use online as well as your hand art skills!!)





Many large companies around the world like The Body Shop. Hyundai and Lacoste have ran massive campaigns for sustainability in order to spread awareness and make a change in our community. Let's take a look at some of the campaigns taken:

- Return, Reuse & Repeat: This campaign was started by The Body Shop where in the brand tied up with TerraCycle, to install product recycling bins in store locations worldwide. It is an ongoing program which motivate customers to return their empty plastic bottles in exchange for a redeemable discount voucher code. Their ongoing hashtag on Instagram has also gone viral #BringBackOurBottles.

- The Disappearance of the Crocodile: This campaign was started by the renowned brand, Lacoste and is related to saving our species worldwide. They are in a 3-year collaboration with IUCN (International Union for Conservation of Nature). The company's winning idea was to boldly replace its iconic crocodile embroidered logo with 10 threatened species and match the number of shirts with the number of individuals left in the wild.



Image of Lacoste shirts with endangered animals.



Image of The Body Shop's packaging for the campaign.

CHAPTER 2 :

SUSTAINABILITY & SPORT

The Sport Effect:

Sport also plays a huge role in sustainability in todays world. In fact, all sports around the world have numerous impacts on our planet.

Sports not only feed our competitive spirit but also are greatly advantageous for our physical and mental health. Some sports such as hiking, swimming and running put man and nature together. However many other sports are not harmonious with the nature around us and have adverse affects on sustainability. Let's take a look at some sports and their impact on our environment.

- **Skydiving, Aerobatics & Other Air Sports:**

These sports are experienced once in a lifetime by us however have you ever noticed, for skydiving a jet takes you up at a high altitude so you can enjoy the thrill of skydiving? And a LOT of fuel is required to do this.

This leads to a heavy Carbon footprint per person per jump! Relatively seen, Skydiving is one of the worst sports for our environment.

Even though we might do it once in our lifetime, there are competitions for skydiving as well in which numerous times these jets go up at high altitudes, burning tons of fossil fuels in our air. Next time you go skydiving, think about this!





- **Golf (It isn't as green as it looks!):**

While you might have seen Golf being played in vast lush green meadows that appear to be preserving the environment by maintaining open land, that's not really true. These vast land spaces have been stolen from massive wildlife forests that were home to diverse species on our planet. The animals that used to live here, are pushed into meadows and forests just for a playground for iron-swinging.

Not only this, Golf courses use a total of 2.08 Gallons of water everyday! Also, for maintaining the rolling green courses, hazardous chemicals and unsafe fertilisers such as pesticides and herbicides which also pollute nearby waters.

Now, Golf doesn't appear to be what you think it was, are we right?







Here are images of lush green gold courses that seem visible to you. But look at the image where the man is spraying pesticides on the grass. This is extremely harmful for the environment and is a pollutant.

- **Motorsport (Formula 1, NASCAR, Drag Racing):**

It is pretty obvious that motorsport will be one of the sports that impact our planet in a very negative manner.

NASCAR vehicles are gas-guzzlers; it is said that every NASCAR weekend, 6,000 gallons of fuel is burned up which is horrible for the environment. NASCAR was labelled the 'A Waste of Gas' by the U.S Government as well.

The same is with Formula 1 and Drag Racing. They also release tons of fuel every race weekend which is extremely bad for the environment and pollutes the air that we breathe.

These sports do translate to huge amount of carbon emissions. Around 120,000 pounds of CO_2 is emitted every race weekend. Motorsport has a large Carbon Footprint!



Formula 1



NASCAR

However, Formula 1 and Drag Racing are moving to a much more eco-friendly future. We hope NASCAR too makes changes and can be environment-friendly as well. Read the next chapter to know more.

ACTIVITY - RESEARCH TIME

What is your favourite sport? Research on its impact on sustainability. Is it helping in a greener environment or does it have negative impacts? Use your research skills and write in the space given below.

Sustainability in Motorsport:

While motorsport does have a huge Carbon Footprint and releases tons of CO2 in the air, it is moving to a much more environment-friendly future. Motorsport is taking big strides in sustainability to make the Earth a healthier place to live in.

Formula E demonstrates a new sustainable era for motorsport and it seems to be an eco-friendly alternative to Formula 1. However, Formula 1 has given us an insight on its cut-edge futuristic technology and does pledge to go net-zero carbon by 2030.

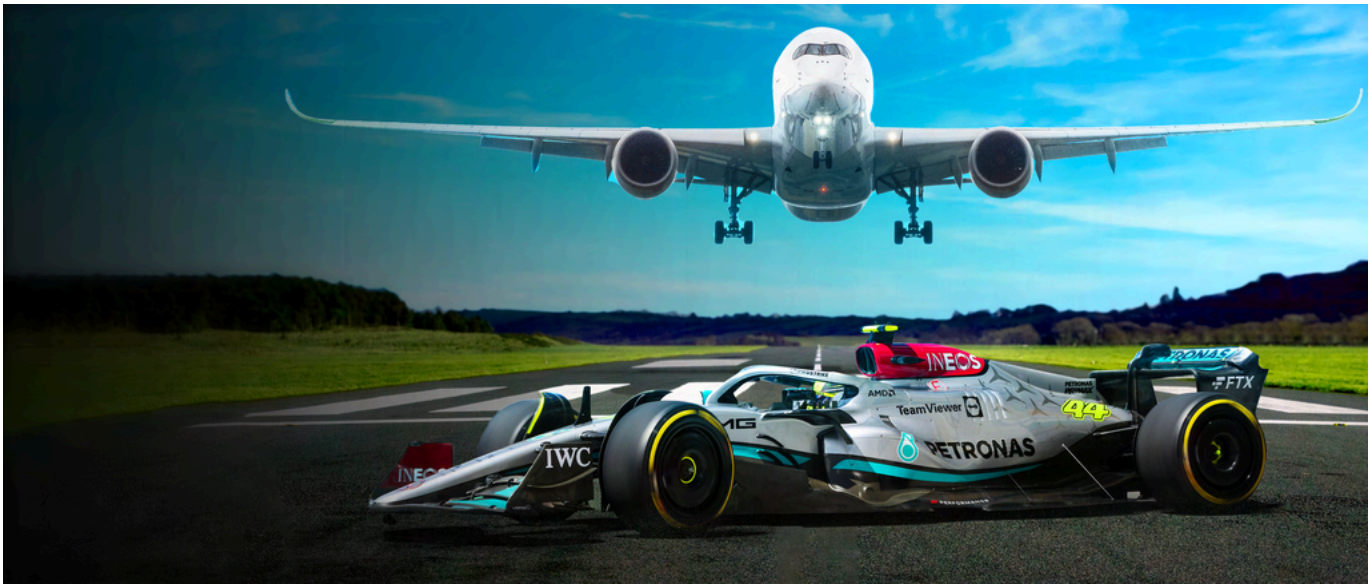
MotoGP is also making efforts in order to make motorsport much more greener and sustainable. MotoGP officials are looking into fuels that will not harm the environment as much.

Let's take a look at what campaigns and changes the motorsport industry is making for a greener surrounding and a more sustainable planet Earth.

Many Formula 1 teams such as Mercedes AMG PETRONAS are moving to a sector called Sustainable Aviation Fuel (SAF). What is SAF?

Sustainable Aviation Fuel is a form or renewable fuel orginally used in air travel. It is recognised as playing a major role in lowering international emissions going forward. SAF is made using sustainable sources such as forestry and agricultural waste, used cooking oil and carbon captured from the air and green hydrogen. SAF recycles more than adding carbon to the environment and hence is sustainable. SAF could contribute to 65% of the reduction in emissions. SAF can also be made without crops in order to prevent deforestation.

Mercedes AMG PETRONAS will be using air planes powered SAF to get to Grand Prix race locations and will reduce air travel footprint by a whopping 50%. This is certainly an amazing campaign by the team for taking Formula 1 to a new level of sustainability.



Not only this, but there are numerous other campaigns by other Formula 1 Racing teams such as Aston Martin. Read ahead to see more interesting campaigns by teams as a result for a sustainable world.

Aston Martin Racing has also taken a campaign in order to make Formula 1 world more sustainable. Aston Martin Racing partnered up with Aramco for F1 2022 Season.

The campaign's aim is to power cars by sustainable fuels by 2025 and be net-zero carbon by 2030. The two enterprises have been using cutting-edge technology and Research & Development to improve engine performance and reduce emissions, contributing to a lower carbon footprint in the sport. The campaign will focus on the production of ultra-efficient hybrid internal combustion engines and advanced fuels, including low carbon synthetic and E-fuels.

This partnership also goes beyond racing, with an ongoing commitment to advance sustainable transport technologies and drive innovation in the global automotive and and transport industries and push to a greener coming future.

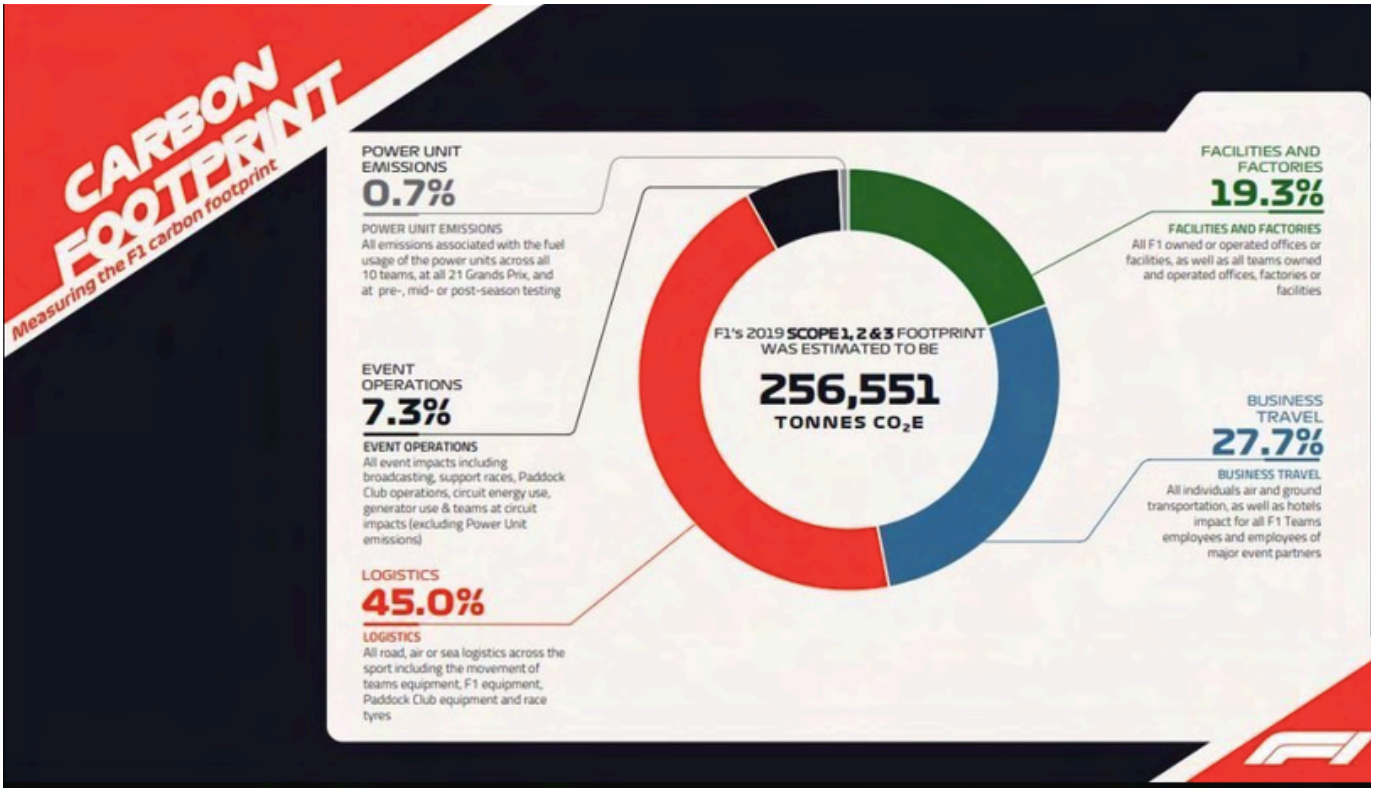




A Brighter Future:

As expected, all sports have been moving closer to a sustainable future. Since they are the ones that use fuel made from renewable resources, which by itself has a significant negative impact on the environment, motorsports must be among the first to see the need to move toward a sustainable future. Fortunately, they are the ones who have advanced the most in the direction of a better and more sustainable future.

The Net Zero Carbon project is being managed by F1 since around 2019. A total of 256000 tonnes of carbon emissions were produced by the cars during that season.



They are attempting to accomplish this goal in the following ways:

Reducing single-use plastics inside the paddock and across the circuit venues has been a simple move, with a focus on encouraging the use of reusable bottles and offering plenty of water stations.

Reusing, recycling, and repurposing race weekend supplies are encouraged. In order to help distribute extra food to those in need in the neighbourhood, several circuits now work with regional food banks and organisations. For instance, after the 2020 Australian Grand Prix was cancelled, approximately 1.5 tonnes of wasted food were given to 17 charities in the state of Victoria.

To provide greener options to travel to events, work is ongoing. Zandvoort is a fantastic example; last year, 25,000 spectators left their automobiles and rode pushbikes to the track, while 40,000 others used public transport or walked there.
Cars will only use sustainable fuels starting in 2026. Advanced sustainable fuels, also known as e-fuels or synthetic fuels, are made up of components derived from a carbon capture system that has been approved, from municipal waste, or from non-food biomass, and achieve a greenhouse gas reduction of at least the level of the current Renewable Energy Directive (RED).



CHAPTER 3 :

SUSTAINABILITY & SCIENCES

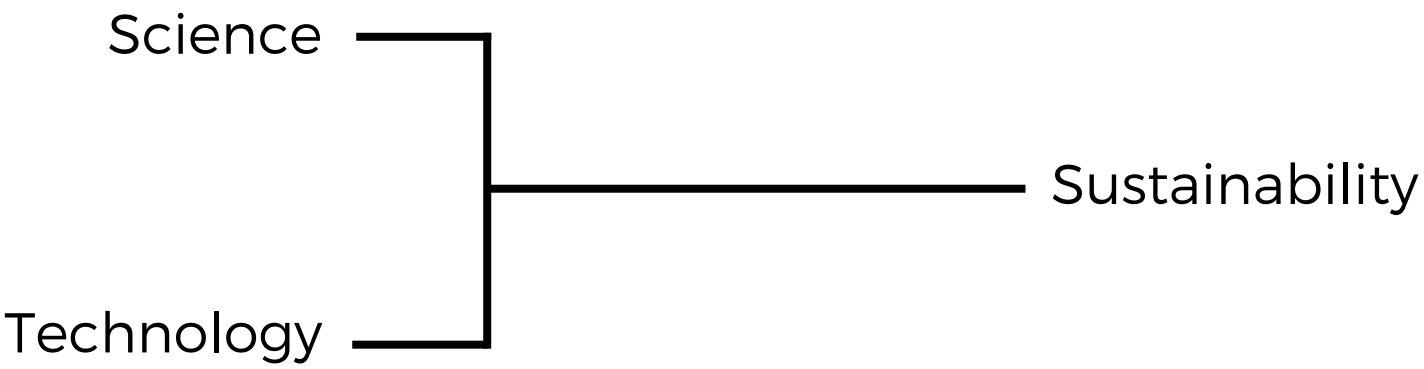
The Negative Effects:

Science and sustainability are also linked together as science also has adverse negative effects on our environment. Science is responsible for new inventions and discoveries on our planet. These make life for humans much easier however do harm the planet in numerous negative ways.

Science and Technology is so important right? Cellphones, laptops, apps, and automobiles are all common examples on how integral technology and science is to our lives. They bring revolutionary improvements to industries and benefit them however many of them also cause environmental damage.

Science and technology has done more damage to the environment more than any other thing. It was science that invented nuclear, chemical and biological weapons that threaten the very existence of our planet.

Science, Technology and the environment are intricately connected, and understanding these connections is critical in the pursuit of sustainability. In this chapter we will discuss various examples of science and technology that have a detrimental influence on sustainability.



- **Heat pollution caused by Technology use:**

As the rate of technological advancement increases, so does heat pollution. The majority of technological devices generate a lot of heat which warms the atmosphere. Automobiles, especially release a great deal of heat and these days even short distances are travelled by motorbikes or scooters.





- **E-waste causes environmental disruption:**

technological waste causes environmental disruption. The majority of technology generates a substantial amount of waste that can be extremely harmful to the environment. Most of the technology used in factories generates a substantial amount of waste that is dumped in lakes or elsewhere, which is extremely hazardous to the environment and to living things. The waste of technology produces toxic gases and odors that are extremely hazardous to life. It also contaminates the water that people use on a daily basis and causes a number of deadly diseases





- **Depletion of Natural Resources:**

Due to the increasing global population, levels of natural resource degradation are also increasing. This has resulted in the estimation of the world's eco-footprint to be one and a half times the ability of the earth to sustainably provide each individual with enough resources that meet their consumption levels. Combined with advancements in technology, development and research, the exploitation of minerals has become easier and humans are therefore digging deeper to access more which has led to many resources entering into a production decline.





- **Nuclear Plants & other forms of energy:**

Non-renewable energy sources like Nuclear Energy are also harmful for the environment. A major environmental concern related to nuclear power is the creation of radioactive wastes such as uranium mill tailings, spent (used) reactor fuel, and other radioactive wastes. These materials can remain radioactive and dangerous to human health for thousands of years. Nuclear plants next to water bodies also release harmful and hazardous toxins that go into the water, contaminating it and sometimes killing marine life organisms inside.





These are some of the many impacts technology and science has on our planet's environment and sustainability. However, in today's world since preserving mankind and our planet has become such a big problem, there has been a complete new branch in the sciences called Environmental Science which deals with the interactions of global, social and human systems as well as concomitant risks to human well-being. Environmental Science provides a platform for building sustainability science as an evolving academic discipline which can point the way to a sustainable global society by facing challenges that existing disciplines have not addressed.

In the 21st century, there have been numerous advancements in science and technology so that we can also preserve mankind and our planet. This means that science and technology is also trying to be much more sustainable. From astrophysics to chemistry and biology, scientists around the world are also discovering new phenomena and inventing new devices in order to build a greener future and continue the human race as we know it.

> The proper use of Science is not to conquer Nature but to live in it
>
> - Barry Commoner

ACTIVITY - RESEARCH TIME

Which is the electronic device you spend the most of your time on? Your laptop, mobile phone, Playstation, Xbox? Research on its impact on our environment. Is it aiding in building a greener future or not? Write down in the space provided below.

Science's & Technology's assistance to Sustainability:

Science... We've all heard of it but how does it contribute to the sustainability of our environment? It creates the groundwork for innovative ideas and solutions, science is essential to addressing humanity's complex concerns, such as climate change, biodiversity loss, pollution, and the decrease of poverty. How do you think humans have achieved the discovery of clean sources of energy? Science!

Science has significantly reduced pollutants in many regions of the world; for instance, new cars are now 99 percent cleaner than they were fifty years ago. The current global energy demand may be met more than 3,000 times by renewable resources. New advances and technology have brought us cleaner, safer water.

Solar panels are just one of the thousand examples of green scientific sustainability. The PV cells on a solar panel capture the energy from the sun's rays as it shines on the panel. In reaction to an internal electrical field within the cell, this energy generates electrical charges that move, which results in the flow of electricity. This brilliant method has allowed the powering of skyscrapers and even small towns!

Technology has also aided the the preserving of our environment. For example, the using of technology to do work such as creating presentations as reduced our reliance on paper which ultimately reduces deforestation!





Sustainable Energy:

Each year the stakes grow higher in the fight to save the environment and combat global warming. Now more than ever, we're aware of the damaging effects that our current dependence on fossil fuels holds for our collective futures. One of the key solutions? Sustainable energy.

Sustainable energy includes any energy source that cannot be depleted and can remain viable forever. It does not need to be renewed or replenished; sustainable energy meets our demand for energy without any risk of going bad or running out. This is why sustainable energy is the answer to our energy needs.

Furthermore, sustainable energy doesn't harm the environment (or at most, there is a minimal risk), increase climate change or cost a heavy price. Although there is a cost associated with creating and building ways to capture sustainable energy, the energy sources themselves are typically free.

Examples of sustainable energy sources include wind, solar and water (hydropower). All of which can be considered inexhaustible and widely available to almost everyone. Geothermal energy can also be included as a sustainable alternative energy source. Geothermal energy creates usable energy from the planet's internal energy sources, such as geysers.



Windmill that uses air for energy



Solar panels that use Sun for energy

Sustainable Energy also brings a lot of benefits along with it. It is not only good for our environment and planet, but also good for YOU! That's right! sustainable energy is also good for us and brings tons of its own benefits. Let's take a deeper dive into sustainable energy's many benefits.



Improves Health:

The burning of fossil fuels in coal and natural gas plants is directly tied with many serious public health issues, such as neurological damage, cancer, heart attacks, breathing problems and premature death. When fossil fuels are burned for energy, they release air and water pollution that negatively affect surrounding communities and environments.

However, all of these associated risks can be eliminated by switching to sustainable energy. Sustainable energy sources typically emit no air or water pollutants that can harm public health. They also don't make a negative impact or strain on local water resources



Decreases your Carbon Footprint:

We're all doing our best to reduce our negative impact on the environment, but there is only so much each person can do. Sustainable energy, such as wind and solar energy, creates zero carbon emissions that can harm the atmosphere and contribute to global warming. It offsets the need for fossil fuels, which are actively increasing climate change and polluting our earth.

Simply switching your own home's energy source can help cut the environmental impact you and your family have on the planet — and eliminate your dependence on fossil fuels



Long-term cost is more affordable:

Sustainable energy can make a huge long-term difference to the electricity and energy costs of our homes and businesses — no matter where you are. Not only are they easily available for little to no cost, but they're much more cost-effective than traditional energy resources, such as power plants.

Making the financial leap to wind, solar or any other sustainable energy source may seem daunting at first because of installation costs, but the long-term investment pays off in just a few years. In fact, sustainable energy costs decrease each and every year. Depending on your country or state, you may even get tax reductions for choosing sustainable energy. Ultimately, you could end up saving exponentially just by making the switch to a sustainable energy system like solar panels.



Energy Security:

Sustainable energy is by definition the most reliable form of energy. It will never deplete and can remove our reliance on imported fossil fuels, which are growing more expensive and less accessible. Sustainable energy also helps to conserve the planet's natural resources and reduce the pollution associated with unsustainable energy.

These are some of the benefits sustainable energy has. But something more important is why sustainable energy is so important for us in the coming future. Let's take a look why.

When we consider sustainable energy and our future, it's essential to also understand what the current energy crisis is doing to our planet right now.

Due to human activity, the earth has grown 1.5 degrees hotter since the industrial revolution — a staggering amount in such a short period of time compared to pre-industrial revolution times. Many factors have contributed to this increase, but a major cause is our burning of fossil fuels for energy. The most prevalent greenhouse gas is carbon dioxide (CO2), which has triggered immense global warming.

Beyond that, finite energy source facilities are polluting the air, water and soil increasingly every day. Besides destroying our natural environments, it's causing undue harm to the health of people globally.

Although humans have done irreversible damage to the environment, there's still hope in how we maintain and mitigate repercussions for future generations. The solution lies in how we stop flooding our atmosphere with carbon dioxide and reduce our emissions.

Remember, sustainable energy produces little to no global warming emissions. By choosing sustainable energy resources and committing to clean energy, humans can effectively slow down the acceleration of climate change.



Future of Science and Sustainability

Here are some innovations which will help to make the future more sustainable

AirCarbon : A Sustainable Plastic For The Future

Newlight Technologies created AirCarbon, which has already won numerous honours for its creative sustainability. It can be used for a variety of purposes and is created from carbon emissions that would otherwise be released into the atmosphere. Since it has been certified to be carbon-negative, the entire production and use process is environmentally friendly and sustainable. It is also a more affordable option to other synthetic materials because, unlike other plastics, it is not derived from oil.



Groasis Waterboxx: Growing trees in the desert is now possible

The Groasis Waterboxx was created by Dutch flower exporter, Pieter Hoff. The Groasis is a planting device that makes growing crops in the desert possible and resource-efficient. It consists of an "intelligent bucket" made from recycled paper, which can germinate seeds, incubate saplings, and water plants. It requires 90% less water than traditional growing methods and can be used in some of the most extreme climates on Earth.



The Veganbottle: An All-Natural Alternative to Plastic Bottles

The Veganbottle, developed by LYSPACKAGING, is made of an all-natural bioplastic that may one day completely replace plastic bottles. The Veganbottle is composed entirely of biodegradable materials, from the cap to the wrapping. Extracts from sugar cane were used to make the bottle. The production of the bottle itself utilises less energy than traditional manufacturing, and sugar cane requires significantly less water than other crops.



CHAPTER 4 :

TEST YOURSELF

Now that you have learnt quite a lot on what sustainability is and how it is important for us and the future of the planet, we have prepared a fun quiz for you to try on.

Sustainability Quiz:

1. **Which of these is a renewable source of energy?**

☐ Fossil Fuels ☐ Sun ☐ Coal

2. **Which of these is NOT a sustainable practice?**

☐ Recycling ☐ Reusing ☐ Littering

3. **Which of the following means the Earth is getting hotter and hotter everyday?**

☐ Climate Change ☐ Global Warming ☐ Thermal Expansion

4. **Which of these is a Non - renewable source of energy?**

☐ Wind ☐ Geothermal ☐ Nuclear

5. **Which type of new fuel is completely sustainable?**

☐ SAF ☐ Petroleum ☐ Diesel

6. **Are sports such as motorsports and golf good for our planet?**

☐ Yes ☐ No ☐ Maybe

7. **Which of the following are one of the benefits of sustainability?**

☐ Human Health ☐ Global Warming ☐ More trees

8. **Which of these is a large problem our century is facing right now?**

☐ Sustainability ☐ Health ☐ Climate Change

9. **Which of these is an element of sustainability?**

☐ Human Health ☐ Plants ☐ Environment

10. **What is SAF made out of making it sustainable?**

☐ Biomass ☐ Plants ☐ Cow Dung

In the previous chapters of this book you have learnt the basics of sustainable energy. Now, take a look at various images and answer if they are renewalbe or non-renewable.

Renewable or Non - Renewable?



Image of Solar Panels (Solar Energy)
Is this Renewable or Non - Renewable?

.....



Image of Nuclear power planet (Nuclear Energy)
Is this Renewable or Non - Renewable?

.....



Image of Geothermal Plant (Geothermal Energy)

Is this Renewable or Non - Renewable? .....



Image of Windmills (Wind Energy)

Is this Renewable or Non - Renewable? .....



Image of Coal (Fossil Fuel)
Is this Renewable or Non - Renewable?

.....



Image of Bioenergy plant (bioenergy)
Is this Renewable or Non - Renewable?

.....

Learning a lot about sustainability through this book, take a look at intriguing facts on sustainability and quotes that might inspire and motivate you in order to create a better planet to live in.

Facts and Quotes!

The USA only makes up 5% of the world's population, but it throws our enough plastic waste to encircle the Earth five times!

Did you know, a modern glass bottle takes up 4,000 or more years to decompose!

The U.S. is the #1 trash-producing country in the world at 1,609 pounds per person per year. This means that 5% of the world's people generate 40% of the world's waste. We toss out two billion plastic razors, a million and a half tons of paper towels, and 12 billion disposable diapers annually!

In the last 170 years, we added 2.4 trillion tons of Carbon Dioxide into our atmosphere. Half of this was added in the last 35 to 50 years!

The five warmest years on record have occurred in the last decade!

Scientists predict that if the increase in greenhouse gas emissions continues unabated, temperatures will rise by as much as 10 degrees Fahrenheit by the end of this century (which is a lot!!)

Quotes

"Sustainability is no longer about doing less harm. It's about doing more good.
- Jochen Zeitz

"Sustainability has to be a way of life to be a way of business"
- Anand Mahindra

"The Earth is what we all have in common. We must protect it"
- Wendell Berry

"We never know the worth of water until the well is dry"
- Thomas Fuller

"The greatest threat to our planet is the belief that someone else will protect it"
- Robert Swan

"Sustainability is treating ourselves and our environment as if we are to live on this planet forever"
- Arron Wood

"Sustainability is about ecology, economy and equity"
- Ralph Bicknese

For our last section of 'Test Yourself' go through the following statements and answer based on whether they are true or false. Circle the answer you think is right.

True or False?

The warmest years are happening now due to multiple reasons like global warming.

TRUE **FALSE**

Humans have added 2.4 trillion tons of CO2 into the atmosphere.

TRUE **FALSE**

The USA, China, Canada, Russia, India and Japan accounted for over 70% of the energy related Carbon Dioxide emission in 2024.

TRUE **FALSE**

Russians are responsible for 30% of global trash.

TRUE **FALSE**

Plastic materials take a couple of days or months to decompose.

TRUE **FALSE**

America uses about 15 times the energy per person than the typical developing country.

TRUE **FALSE**

Lighting accounts for 15% of the global electricity use.

TRUE **FALSE**

The population has increased more than twice the water use.

TRUE **FALSE**

Nearly half of the solid waste produced globally is organic or bio degradable.

TRUE **FALSE**

REFLECTION

As a last task of this book, we'd like you to write a small reflection of what you have learned from this book and what you think you can do for a sustainable future. Write in the space provided below.

CHAPTER 5 :

SUPPORT US

We highly appreciate your purchase of this book. We hope you found it informative, fun and knowledgeable as well as learnt tons of what sustainability is in basic.

At Infinity Racing, we believe that sustainability plays a major role in our team and at the F1 In Schools Competition. We believe in a greener future and the main concept of environmental sustainability.

As we embark on one of our most ambitious campaigns yet, we'd love each and every one of your support. Wish us all the best for the F1 In Schools World Finals 2023 as we represent our country at the highest and most challenging level of the competition. However that's not all!

By supporting us, we'd also like you to support the many sustainable campaigns we hold as a team across our journey in Formula 1 In Schools. For further details on the campaigns we hold, view our Social Media pages (Instagram, Twitter, Facebook, YouTube). If possible, do connect with us in any way possible if you'd like to support us in any campaigns. Furthermore, do put our social media posts on your platforms in order to spread awareness on such a serious matter that is affecting our entire planet Earth. We encourage you to also run campaigns along with your school or society to raise awareness and make a change.

For contacting us for any additional details:



@f1s.infinity



@f1Infinity_in

We hope to see absolutely amazing campaigns relating to sustainability by you, reading this book. Let us know if we can help you in any way possible.

Till then, Signing off,
Infinity Racing

INFINITY RACING

ESTD. 2022

INFINITE PLANET

INFINITY RACING

India's F1 in Schools World Finalists, Infinity Racing, wrote the book Infinite Planet. You will discover many things about the globe through this book that you probably did not know previously. This book provides information on a wide range of topics relating to sustainability, including science, motorsport, the future, and a host of other topics. If you are up to the challenge, there are also enjoyable activities scattered throughout the book to test your knowledge.

